Rolling resistance

LAMMPS dem07 | Rolling torque and angular-velocity decay | 28 October 2026

PASS	All four runs pass the official 1% spin check at 7 ms.

Official spin check at 7 ms

Run	dt (us)	omega at 7 ms	Error (%)	omega at 10 ms	1% check
CPU dt	1	0.64994130	0.00903	0.50004671	PASS
GPU dt	1	0.64994130	0.00903	0.50004671	PASS
CPU dt/2	0.5	0.64992856	0.01099	0.50003399	PASS
GPU dt/2	0.5	0.64992856	0.01099	0.50003399	PASS

Angular velocities are in rad/s. The official rolling_decay criterion tests omega_y = 0.65 at 7 ms with 1% relative tolerance. The ideal final value 0.5 at 10 ms is supplementary. Inputs and criteria were retained after inspecting results.

Inputs and direct parameter mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	Diameter 0.005 m; density 2700 kg/m^3; center (0, 0, 0.002499982) m; v = 0; omega_y = 1 rad/s
Wall / gravity	Fixed plane z = 0; g = 1 m/s^2 downward; periodic x/y immaterial for the isolated stationary sphere
Normal contact / damping	Hooke; kn = 10000 N/m; restitution e = 1; automatic contact damping = 0
Sliding / rolling	Sliding stiffness, damping and friction = 0; rolling spring kr = 10000 N/m; rolling damping = 0; mu_r = 0.05
Other resistance	Torsional friction and global damping = 0
Elastic placeholders	Official Hooke input gives no E or Poisson ratio. FunDEM E = 1e4 Pa and nu = 0.33 are inactive: kn, ks, kr and ktorsion are explicitly 10000, 0, 10000 and 0 N/m.
Time / saved output	10 ms; dt = 1 / 0.5 us; 10000 / 20000 steps; 1001 native frames per run at 10 us; every-step CSV

Analytic decay and instantaneous torque

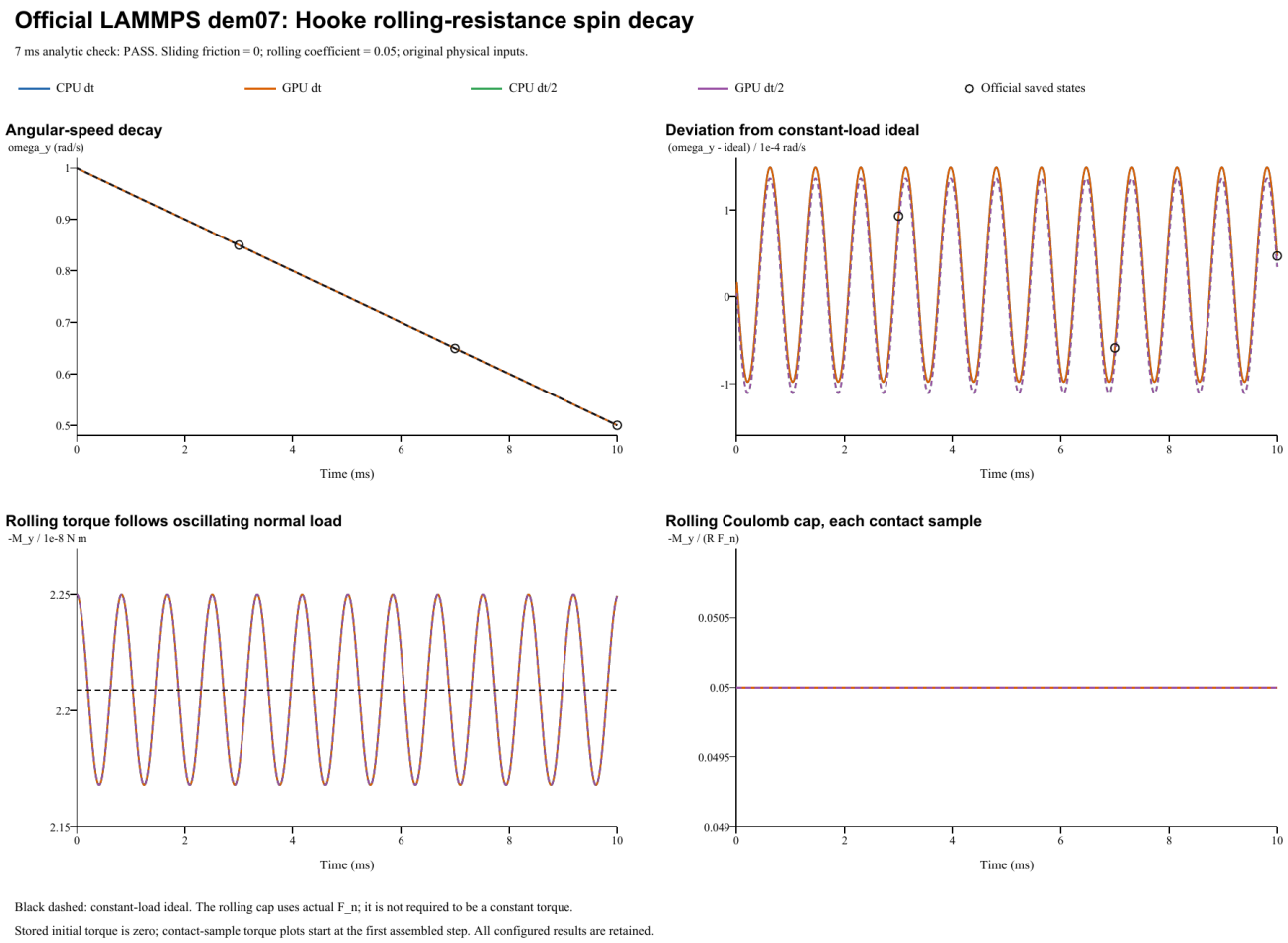
I = (2/5) m R^2 = 4.417864669e-10 kg m^2.
M_ideal = mu_r R mg = 2.208932335e-08 N m.
d(omega_y)/dt = -50 rad/s^2; omega_y(t) = 1 - 50 t.
At 3 / 7 / 10 ms the ideal values are 0.85 / 0.65 / 0.5 rad/s.

Reference provenance

Official YAML and golden states; analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML Git blob 70c8d8237edc01e7b207a9a1724b416118e179b0. LAMMPS was not rerun locally. YAML epsilon 1e-10 is not a universal cross-code tolerance. The official check is at cumulative step 7000, not the 10000-step endpoint.

Curves and retained evidence

Native angular motion, torque, instantaneous load and reference checkpoints



After the initial zero-torque state, native resisting torque follows -mu_r R F_n using the instantaneous normal force. The small initial height offset causes about 1.86% normal-force oscillation; the constant-load angular-speed line is an approximation. The initial rolling history is zero. Full-step angular velocity agrees with the trapezoidal integral of rolling torque. At 7 ms, dt refinement need not monotonically reduce distance to the constant-load target because normal-load oscillations remain physical. CPU/GPU spin histories agree exactly. Max. torque-coefficient error is 6.94e-18; angular-impulse residual is 1.1e-16 rad/s. Halving dt changes omega_y at 7 ms by 1.27e-05 rad/s. Rolling uses R; the dem05 force arm R-delta is inactive because Ft = 0.

Complete native results are retained

CPU dt: open run.json; 10,000 steps / 1,001 native frames.

GPU dt: open run.json; 10,000 steps / 1,001 native frames.

CPU dt/2: open run.json; 20,000 steps / 1,001 native frames.

GPU dt/2: open run.json; 20,000 steps / 1,001 native frames.

Run manifest | Verified analysis | Checkpoint CSV | Contact audit | Full native archive ZIP. Native run.json, project.json, frames/contacts, input projects, audits, backend logs and per-step CSVs are retained.

Native output is every 10 us including both endpoints: 1001 frames per run, every 10 base steps or 20 half steps. Every configured frame is retained; each integration step is separately recorded in CSV.

Execution and scope

Real FunDEM SimulationSession runs used linked CPU/CUDA native libraries. No solver or input parameter was adjusted to force agreement. CPU/GPU agreement and timestep refinement are separate numerical evidence; the official 7 ms check determines pass/fail. GUI executable byte identity was not established.

This case checks resistance to rotation about a tangential axis and angular-velocity decay. Sliding friction is zero. Torsional resistance, translating rolling motion and many-particle flows require other tests.